EcoCycle: A Sustainable Wastewater Filtration System

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Abdelrahman Ashraf, Ahmed Elshafey, Ahmed Magdy, Hamza Khaled

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Abstract

This project addresses major Grand Challenges such as ensuring access to clean water and promoting sustainable energy generation, with a focus on mitigating agricultural wastewater pollution in Egypt. Rural areas face growing environmental and public health risks due to untreated wastewater, compounded by limited access to affordable energy. The proposed solution is a self-powered, three-stage vertical filtration prototype made from recycled materials and driven by renewable mechanical energy. The design aims to meet essential water quality standards, including pH near 7.00, turbidity between 1 and 3 NTU, and TDS below 500 ppm. The first filtration stage performs physical treatment using polyester to remove suspended solids and a chlorine-soaked cotton layer for biological disinfection. The second stage introduces chemical treatment through a mixture of ferric sulphate and limestone, enhancing coagulation and stabilizing the water's pH. The final stage uses 100 ± 0.01 g of zeolite for ion exchange, effectively targeting salt removal. The system is automated with pH, TDS, and turbidity sensors feeding real-time data to an Arduino controller, which reprocesses water as needed and transmits updates via Adafruit IO. Following sensor calibration, two test plans were conducted. The first test failed to meet TDS targets due to ineffective zeolite performance. After optimization, the second test achieved significant improvements: pH 6.99 ± 0.05, TDS 415 ppm ± 4%, and turbidity 5.70 ± 0.12 NTU, successfully meeting most design requirements. The system demonstrated an average filtration efficiency of 79.27% ± 5.56%, and low energy consumption between 2.56 and 5.89 kJ per cycle, supporting its viability as a sustainable and scalable water treatment solution.

Introduction

The lack of clean water access stands as a critical problem in Egypt's rural and agricultural areas because wastewater flows directly into unprocessed state into the environment. Agricultural wastewater consists of multiple harmful elements, such as suspended solids, together with organic matter and microorganisms, and high salinity, which lead to deterioration of soil and water quality. The project directly addresses the national challenges identified in Egypt by focusing on pollution reduction and clean water resource expansion, and sustainable agricultural support. The main issue to be addressed involves water source contamination through untreated agricultural runoff that harms crop production and food quality standards while creating significant health risks for the public. Learning from previous solutions will help overcome these challenges.

The Bahr El-Baqar Wastewater Treatment Plant ***(I Azzam, 2022),*** the world's largest, boosts agriculture, saves water, and reduces pollution. Despite challenges like soil instability and skill gaps, it fosters economic growth, job creation, and sustainable development in Egypt's arid regions. The system filters water in three phases using various materials: polyester membranes (10-micron) in 120-disc filters, chemicals like sulfuric acid, sodium hydroxide, ferrous chloride, and polymers for flocculation and pH adjustment, and ozone or chlorine for sterilization—ensuring effective removal of solids, pathogens, and organic contaminants for irrigation-quality output. Therefore, defining clear design requirements became necessary to guide the development of a more effective and well-structured solution.

The life application of the solution is to use the treated water for domestic use. The design requirements of this application are to reach pH near 7.00, turbidity between 1 and 3 NTU, and TDS below 500 ppm. To achieve these design requirements, a complex filter must be available.

A three-stage closed-loop water purification system with natural and recycled materials will serve as the solution for treating agricultural wastewater. The built prototype consists of three vertically stacked containers that execute different purification functions. Physical and biological filtration with polyester fabric and cotton, as well as chlorine treatment, occur at the first stage to eliminate large particles and microorganisms. The second stage implements chemical coagulation with ferric sulphate alongside pH balancing through limestone addition, before the third stage uses zeolite for salinity reduction through ion exchange. Real-time measurements enable the system to operate solenoid valves and pumps for repeated purification cycles. The system presents an efficient, sustainable solution to water scarcity and pollution problems in Egyptian rural areas through innovative engineering methods.

Materials

Table (1): Shows the materials

Item	Description	Quantity	Cost (EGP)	Picture
Wooden base	The base holds other materials	1	100	
Containers	Hold the filter to purify water	3	45	
Lime	The second filter to make coagulation	0.2 g	5	
Zeolite	The third filter make ion exchange	100 g	140	
5V-Pump	Move water from stage to the second	3	165	
12V-Valve	Stop the water flow until the filter time end	2	380	
Battery	Turn pumps and valves	12	420	
Relays	Controlling High-Voltage Devices	5	175	
Arduino Uno	Control the system	1	400	
pH Sensor	Calculate the pH value	1	1750	
TDS Sensor	Calculate the salinity value in ppm	1	900	
Turbidity Sensor	Calculate the Turbidity value in NTU	1	800	
Total Cost = 5280 EGP				

Methods

The prototype was built on a wooden base with dimensions of 82 cm in height, 18 cm in length, and 14 cm in width (±0.5 cm). Three transparent 1.5-liter plastic containers housed the filtration stages, vertically aligned and fixed to the base using wax for stability and sealed internal components, as shown in Figure (1). The filtration process was divided into three stages ***(ES 2.05):*** The first stage consisted of a polyester fabric layer above a cotton layer saturated with 7.5 ± 0.1 mL of chlorine. The polyester filtered large, suspended solids, while the chlorine-soaked cotton acted as a disinfectant, eliminating microorganisms. The second container contained a 0.2 ± 0.01 g mixture of ferric sulfate and limestone ***(CH 2.08),*** enclosed in a polyester fabric pocket. Ferric sulfate acted as a coagulant to remove finer particles, while lime stabilized the pH of the water. The third stage utilized 100 ± 0.01 g of zeolite, enclosed in a polyester fabric pocket, for salt removal through ion exchange, balancing the salinity. ***(Kumari et al., 2024)***

Figure (1): Shows the construction of the prototype

Before testing, the sensors pH, TDS, and turbidity were calibrated: the pH sensor with standard buffer solutions pH 4.00, 7.00, and 10.00, the TDS sensor with distilled water 0 ppm, and the turbidity sensor with distilled water 0.00 NTU.

The system was built with two solenoid valves and three water pumps, connected to an Arduino Uno via relays as shown in Figure (2). The relays ensured safe operation by stepping down the voltage to levels compatible with the Arduino. The solenoid valves were placed at the outlet of each filtration stage and remained closed until activated by pressure from the pumps.

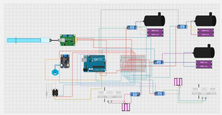

Figure (2): Shows the schematic diagram

Sensors were integrated to automate the process. The pH, TDS, and turbidity sensors were positioned at both the inlet and outlet of the filtration cycle. The data from these sensors was processed by the Arduino in real-time. If the post-cycle readings did not meet the target values, the Arduino would automatically restart the treatment loop. The system also featured a mobile dashboard created with Adafruit IO for live monitoring, with sensor data logged and transmitted wirelessly.

The test phase began by filling the first container with untreated agricultural wastewater. The five-cycle filtration process lasted approximately 15 minutes per cycle ***(BI 2.09).*** Each cycle began with 30 seconds of baseline sensor measurements. Then, the first pump activated for 30 seconds, pushing the water into the first filtration stage. The water passed by gravity to the second stage, with the outlet valve of the second container remaining closed. The water stayed in the second stage for 7 minutes—2 minutes for sedimentation and 5 minutes for coagulation and pH stabilization through the ferric sulfate and limestone.

Following this, the first solenoid valve and second pump were triggered for 1 minute, moving the water into the third stage with zeolite. The system paused for 5 minutes for ion exchange and salinity control. The second solenoid valve and third pump were activated for 1 minute to discharge treated water into the final collection tank. A third set of sensor readings was taken to conclude the cycle. If the readings met the target values, the system proceeded to the next cycle.

Initially, after the first cycle, TDS levels remained above the acceptable threshold. To address this, the zeolite layer in the third stage was optimized for better salt absorption. The target treatment standards were pH close to 7.00, TDS below 500 ppm, and turbidity between 1 and 3 NTU, ensuring that the treated water was safe for domestic use.

Results

As shown in Table (2) and Figures (3–5) present the sensor data collected during two test plans conducted to evaluate the filtration system's effectiveness. The first test showed partial improvement but ultimately failed to meet the treatment targets, particularly for Total Dissolved Solids (TDS), which dropped only from 1113 ± 4% ppm to 768 ± 4% ppm, remaining well above the acceptable threshold. The turbidity decreased from 10.49 ± 0.12 NTU to 5.92 ± 0.12 NTU, and pH rose from 5.14 ± 0.05 to 6.92 ± 0.05, approaching neutrality. Despite the observed improvements, the filtration system could not fully remove salts due to the insufficient performance of the zeolite layer in the third stage. This highlighted the need for further adjustments and optimization of the ion exchange medium.

In contrast, the second test plan succeeded in meeting the targets, with final TDS reduced to 415 ppm ± 4%, pH stabilized at 6.99 ± 0.05, and turbidity lowered to 5.70 ± 0.12 NTU. The graphical data in Figures (3-5) confirms consistent improvement across all parameters through five cycles. These results validate the prototype's potential once appropriately tuned.

Table (2): Representing the test plans' results

Test plans	Turbidity Before	Turbidity After	pH Before	pH After	TDS Before	TDS After	Statue
First	10.49 ± 0.12	5.92 ± 0.12	5.14 ± 0.05	6.92 ± 0.05	1113 ppm ± 4%	768 ppm ± 4%	Failed
Second	10.05 ± 0.12	5.70 ± 0.12	5.09 ± 0.05	6.99 ± 0.05	1165 ppm ± 4%	415 ppm ± 4%	Succeeded

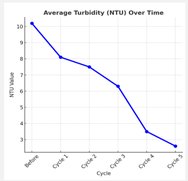

Figure (3): Shows the Turbidity Sensor reading through the 5 cycles

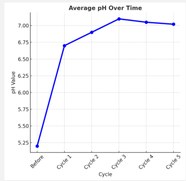

Figure (4): Shows the pH Sensor reading through the 5 cycles

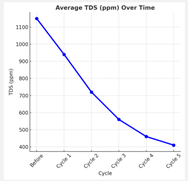

Figure (5): Shows the TDS Sensor reading through the 5 cycles

Analysis

Agricultural wastewater has been identified as a major contributor to environmental degradation and water contamination in Egypt. Its untreated discharge increases salinity, promotes pathogen growth, and introduces harmful ions such as ammonium (NH_4^+) into freshwater systems. The ingestion of nitrates resulting from ammonium oxidation has been linked to methemoglobinemia (Blue Baby Syndrome), particularly in infants. The proposed system addresses these issues by targeting salinity reduction, biological contamination, and suspended solids removal, thereby contributing to improved water quality, environmental health, and public safety. ***(Abd Ellah, 2020)***

The prototype functions through three sequential treatment stages:

Primary Treatment: Physical filtration is achieved using polyester fabric to remove large, suspended solids. Chemical disinfection is carried out using chlorine ***(Yuthawong et al., 2024),*** which reacts with water to form hypochlorous acid (HOCl), a potent antimicrobial agent: $Cl_2 + H_2O \rightarrow HCl + HOCl$

Secondary Treatment: This stage incorporates coagulation and pH adjustment ***(Sulistyo et al., 2012).*** Ferric sulfate ($Fe_2(SO_4)_3$) facilitates the removal of fine particles by forming ferric hydroxide precipitates: $Fe_2(SO_4)_3 + 6H_2O \rightarrow 2Fe(OH)_3\downarrow + 3H_2SO_4$, Simultaneously, lime ($Ca(OH)_2$) is used to neutralize excess acidity: $Ca(OH)_2 + H_2SO_4 \rightarrow CaSO_4 + 2H_2O$

Tertiary Treatment: The final stage involves ion exchange using natural zeolite ($Na_2Al_2Si_3O_{10}\cdot 2H_2O$). Zeolite removes hardness-causing ions such as calcium (Ca^{2+}) and ammonium (NH_4^+) by replacing them with sodium ions: $Na\text{-}Zeolite + Ca^{2+} \rightarrow Ca\text{-}Zeolite + 2Na^+$

pH Sensor: Calibrated with buffer solutions pH 4, 7, 10. Voltage readings were mapped to pH values in code.
TDS Sensor: Calibrated using distilled water (0 ppm) and adjusted via code.
Turbidity Sensor: Set to 0.00 NTU using distilled water; voltage mapped in code.

Efficiency Calculation Formula: **Efficiency = ((Initial Value - Final Value) / Initial Value) × 100**
For Example: Initial Turbidity = 10.146 NTU, Final Turbidity = 2.474 NTU
Efficiency = (10.146 - 2.474) / 10.146 × 100
Efficiency = 7.672 / 10.146 × 100 ≈ 75.6%
Average Efficiency Calculation Formula:
Average Efficiency = (Efficiency$_1$ + Efficiency$_2$ + Efficiency$_3$) / 3
= (75.6 + 37.97 + 64.4) / 3 = 177.87 / 3 ≈ 59.3%
Error Calculation Formula:
Error (%) = (Measurement Uncertainty / Measurement Value) × 100

Table (3): Shows Each Stage Efficiency and the Average Efficiency

Stage	Efficiency
First	75.6% ± 3.13%
Second	97.8% ± 5.16%
Third	64.3% ± 8.4%
Average	79.27% ± 5.56%

The initial filtration test demonstrated limited effectiveness in reducing salinity, with TDS levels decreasing from 1113 ± 4% ppm to 768 ± 4% ppm, resulting in an efficiency of 31.0% ± 4.9%. This low efficiency highlighted the need for optimization.
Following this, the amount of zeolite was increased from 50 ± 0.01 g to 100 ± 0.01 g, which significantly improved performance as shown in Figure (6). In the adjusted setup, TDS levels dropped from 1165 ± 4% ppm to 415 ± 4% ppm, achieving an enhanced efficiency of 64.3% ± 8.4%.

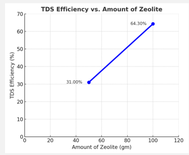

Figure (6): Shows the change in efficiency by changing the amount of zeolite

Water volume and TDS levels were measured every 10 minutes over 70 minutes to evaluate the prototype's performance.

The volume of water gradually decreased from 1500 ± 2% mL to 1340 ± 2% mL. The loss followed a consistent pattern, with minor fluctuations due to rural testing conditions and manual timing.

The TDS level decreased over five filtration cycles, dropping from 250 ± 4% ppm to 50 ± 4% ppm, with a reduction of approximately 50 ± 4% ppm per cycle. This consistent degradation indicates effective filtration, though efficiency naturally tapers as the media approaches saturation.

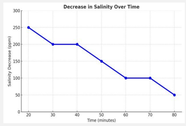

Figure (7): Shows the decrease in salinity for each 10 minutes

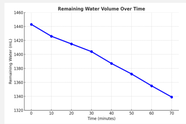

Figure (8): Shows the decrease in volume for each 10 minutes

As shown in Figure (9), the lifetime of each filtration stage is illustrated, allowing clear identification of when each component should be replaced. The final 10% of each stage's lifespan is highlighted in red to indicate the critical threshold. Additionally, the average lifetime is calculated to guide periodic maintenance scheduling.

As shown in Table (4), the energy used by each component was calculated ***(ME 2.05)*** using the formula **P = IVt.** The total energy consumed per cycle from 2.56 to 5.89 kJ, and the total energy used over five cycles from approximately 12.80 to 29.45 kJ.

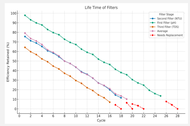

Figure (9): Shows the lifetime of the three filters and average efficiency

Table (4): Shows the energy consumption in one

Item	Current	Voltage	Time	Power= I.V.T
pH Sensor	10 mA	5 V	30 s	1.5 J
TDS Sensor	3-6 mA	3.3-5 V	30 s	0.297-0.9 J
Turbidity Sensor	20 mA	5 V	30 s	3 J
Arduino Uno R3	50-80 mA	5 V	900 s	225-360 J
Water pump	200-460 mA	5 V	150 s	150-345 J
Solenoid valve	0.42-2.5 A	12 V	120 s	604.8-3600 J
Relay	70mA	5 V	4500 s	1575 J
Total energy consumed = 2559.6-5885.4 J				

As shown in Figure (10), the mobile app interface displays real-time graphs of pH, TDS, and turbidity, with a power switch and status light. Data is stored when active, and green indicates successful treatment. ***(CS 2.05)***

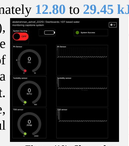

Figure (10): Shows the mobile application interface

Learning transfer:

Table (5): Shows the beneficial L.Os

L.O	Description
BI 2.09	Like hormonal feedback maintains balance, the project uses sensor-driven feedback loops to regulate water purity for sustainable reuse.
CH 2.08	Electrolysis causes ion precipitation, like ferric sulphate, forming flocs that trap impurities, aiding coagulation in wastewater treatment stages.
CS 2.05	The data layer collects sensor readings, stores values in a database, and the logic layer analyzes trends for water quality control.
ES 2.05	Like Earth's crust formed through cooling, layering, and mineral exchange, the project filters water by screening, coagulation, and ion exchange.
ME 2.05	Mechanical power measures energy used by pumps, helping calculate system efficiency and energy cost per liter of treated wastewater.

Conclusion

The results of the filtration prototype demonstrate that a sustainable system can effectively improve water quality using recycled materials. This solution was developed in response to Egypt's growing water pollution issues, particularly in regions like Bahr El-Baqar, where agricultural runoff has severely impacted freshwater sources. The prototype employs a three-stage treatment process—physical filtration, chemical coagulation, and ion exchange.

After optimization of the zeolite layer, the second test confirmed that the prototype could meet key design targets, with final values of pH 6.99 ± 0.05, TDS 415 ppm ± 4%, and turbidity 5.70 ± 0.12 NTU. These outcomes validate the system's ability to treat wastewater to near-safe standards. Additionally, it achieved an average filtration efficiency of 79.27% ± 5.56% while consuming only 2.56 to 5.89 kJ per cycle, highlighting its potential for use in off-grid, rural communities.

In comparison with centralized systems, this prototype offers a decentralized, affordable alternative for areas like Bahr El-Baqar. It supports both environmental sustainability and public health, making it a viable model for future community-scale water treatment solutions.

Recommendations

The following recommendations aim to boost system performance, speed, and long-term effectiveness based on prototype results and testing outcomes:

- **Natural Resins Instead of Zeolite:** Replacing zeolite with natural plant-based ion-exchange resins improves salinity reduction. These resins have higher selectivity for sodium and chloride ions, faster regeneration, and longer usability. This results in more efficient and sustainable water treatment.
- **Electric Mixing Motors in Stages 2 & 3:** Installing electric motors with stirring blades in the chemical and ion-exchange stages enhances mixing and reaction rates. It shortens treatment time, ensures uniform distribution of reactants, and increases overall efficiency per cycle.
- **Redesigned Container Layout for Better Flow:** Modifying the container layout with angled or funnel bases improves gravitational flow and reduces retention time. It prevents outlet clogs, increases flow speed, and minimizes the need for manual adjustments.

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For Further Info

Abdelrahman Ashraf Elnakesh, abdelarhman.2223029@stemsharkya.moe.edu.eg
Ahmed Anwar Elshafey, ahmed.2223001@stemsharkya.moe.edu.eg
Ahmed Magdy Attia, ahmed.2223006@stemsharkya.moe.edu.eg
Hamza Khaled Abdelrahman, hamza.2223020@stemsharkya.moe.edu.eg